

Subject Description Form

Subject Code	ME34002
Subject Title	Engineering Thermodynamics
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AP10005 Physics I, AMA2111 Mathematics I, and ME34005 Fundamentals of Thermal Science and Engineering (subj. to approval) Exclusion: ME34001 Engineering Thermodynamics
Objectives	To extend students' knowledge of fundamental of thermodynamics to various energy conversion systems, including combustion processes, steam and gas power cycles, refrigeration cycle, air-conditioning and fuel cells.
Intended Learning Outcomes	a. Understand the fundamental principles of thermodynamic systems. b. Formulate and solve thermodynamic problems relating to combustion, steam power, gas power and refrigeration cycles, and air-conditioning. c. Analyze and interpret data obtained from experiments in engineering thermodynamics. d. Present effectively in completing written reports of laboratory work and the given task.
Subject Synopsis/ Indicative Syllabus	<i>Review of Basic Concepts of Thermodynamics</i> - First and second law of thermodynamics. Non-flow and steady-flow processes. Carnot cycle. Entropy. <i>Power Cycles</i> - Vapour cycles: Rankine cycle. Superheat and reheat. Air standard engine cycles: Otto cycle and Diesel cycle. Gas turbine cycles. <i>Refrigeration Cycles</i> - Refrigerator and heat pump. Vapor compression cycle. Coefficient of Performance. <i>Thermodynamic Property Relations</i> – The Maxwell Relations. The Clapeyron Equation. General Relation for change of internal energy, enthalpy, entropy and specific heat. The Joule-Thomson Coefficient. <i>Psychrometry and Air Conditioning</i> – Gas-vapor mixture. Psychrometry. Psychrometric chart. Air conditioning processes. <i>Combustion</i> - Hydrocarbon fuels. Combustion equations. Air fuel ratio. Lean and rich mixture. Theoretical and actual combustion processes. Enthalpy of combustion. Adiabatic flame temperature. <i>Fuel Cells</i> - Basic principle. Thermodynamics and reaction kinetics. Polarization. Water and thermal management. Laboratory Experiment There are two 2-hour laboratory sessions with the typical experiments: 1. Refrigeration system 2. Combustion

Teaching/Learning Methodology	Lectures are used to deliver the fundamental knowledge in relation to thermodynamics and heat transfer (outcomes a and b). Tutorials are used to illustrate the application of fundamental knowledge to practical situations (outcomes a, b and d). Experiments are used to relate the concepts to practical applications and students are exposed to hand-on experience, proper use of equipment and application of analytical skills on interpreting experimental results (outcomes c and d). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>Experiment</td><td>√</td><td></td><td>√</td><td>√</td></tr></table>						Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√			Tutorial	√	√			Experiment	√		√	√																
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Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Examination</td><td>60%</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>2. Tests</td><td>20%</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>3. Homework/Classwork</td><td>10%</td><td>√</td><td>√</td><td></td><td></td></tr><tr><td>4. Laboratory Work/Reports</td><td>10%</td><td></td><td></td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.6 × End of Subject Examination + 0.4 × Continuous Assessment Examination is adopted to assess students on the overall understanding and the ability of applying the concepts and knowledge. It is supplemented by the tests (usually two), design project (and report) and laboratory work (and reports) which provide timely feedbacks to both lecturers and students on various topics of the syllabus.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Examination	60%	√	√			2. Tests	20%	√	√			3. Homework/Classwork	10%	√	√			4. Laboratory Work/Reports	10%			√	√	Total	100%				
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Reading List and References	1. Y.A. Cengel, M. A. Boles, M. Kanoglu. Thermodynamics: An Engineering Approach, McGraw Hill, latest edition, in SI units. 2. R.E. Sonntag, C. Borgnakke and G.J.V. Wylen, Fundamentals of Thermodynamics, John Wiley and Son, latest edition. 3. T.D. Eastop and A. McConkey, Applied Thermodynamics for Engineering Technologists, Pearson, latest edition. 4. K. Wark, and D. Richards, Thermodynamics, McGraw-Hill, latest edition. 5. K.D. Hagen, Heat Transfer with Applications, Prentice Hall, latest edition. 6. F.D. Incropera, and D.P. Dewitt, Introduction to Heat Transfer, Wiley, latest edition.
Last Update	August 2024